Partially Asynchronous Optimization on Manifolds

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I. Simulations

A. Background on Unit Quaternions

Let $q \in S^3$ be a unit quaternion, given by

$$q = q_w + q_x i + q_y j + q_z k = q_r + q_v \in S^3.$$

The conjugate of q is denoted q^* and given by

$$q^* = q_w - q_x i - q_y j - q_z k = q_r - q_v.$$

It holds that $(q^*)^* = q$, and $(pq)^* = q^* p^*$.

The quaternion product is given by

$$\begin{aligned} pq &= (p_r q_r - p_v \cdot q_v) + (p_r q_v + q_r p_v + p_v \times q_v) \\ &= (p \cdot q^*) + (p_r q_v + q_r p_v + p_v \times q_v). \end{aligned}$$

A rotation of θ radians around the axis $u \in \mathbb{R}^3$ is given by

$$q = \cos\frac{\theta}{2} + u \sin\frac{\theta}{2}.$$

Represent a vector $v_v \in \mathbb{R}^3$ as a pure quaternion, $v = (0, v_v)$. The rotation of a vector $v_v \in \mathbb{R}^3$ by a unit quaternion q is given by

$$L_q(v) = qvq^*.$$

It follows that the composition of L_{q2} and L_{q1} is given by

$$L_{q2}(L_{q1}(v)) = L_{q2\,q1}(v).$$

Equivalently, the rotation of a vector by a quaternion can written using rotation matrices. Let $R(q) \in SO(3)$ be the rotation matrix induced by q, such that

$$L_q(v) = R(q)v_v,$$

where

$$R(q) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_z q_w) & 2(q_x q_z + q_y q_w) \\ 2(q_x q_y + q_z q_w) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_x q_w) \\ 2(q_x q_z - q_y q_w) & 2(q_y q_z + q_x q_w) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}.$$

This representation can be differentiated with respect to q_w, q_x, q_y, and q_z, via

$$\begin{aligned} \frac{\partial R(q)}{\partial q_w} &= \begin{bmatrix} 0 & -2q_z & 2q_y \\ 2q_z & 0 & -2q_x \\ -2q_y & 2q_x & 0 \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_x} &= \begin{bmatrix} 0 & 2q_y & 2q_z \\ 2q_y & -4q_x & -2q_w \\ 2q_z & 2q_w & -4q_x \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_y} &= \begin{bmatrix} -4q_y & 2q_x & 2q_w \\ 2q_x & 0 & 2q_z \\ -2q_w & 2q_z & -4q_y \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_z} &= \begin{bmatrix} -4q_z & -2q_w & 2q_x \\ 2q_w & -4q_z & 2q_y \\ 2q_x & 2q_y & 0 \end{bmatrix}. \end{aligned}$$

Let f be some function of $R(q)$, and let $\nabla_{R(q)} f$ denote the gradient of f with respect to $R(q)$. Then, the gradient of f with respect to q is given by

$$\nabla_q f = \begin{bmatrix} \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_w} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_x} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_y} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_z} \rangle_F \end{bmatrix}.$$

Let $c \in \{w, x, y, z\}$. For clarity, we can see

$$\begin{aligned}
\left\langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_c} \right\rangle_F &= \left\langle \begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{13}} \\ \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{23}} \\ \frac{\partial f}{\partial R(q)_{31}} & \frac{\partial f}{\partial R(q)_{32}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix}, \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right\rangle_F \\
&= \mathrm{Tr}\left(\begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{13}} \\ \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{23}} \\ \frac{\partial f}{\partial R(q)_{31}} & \frac{\partial f}{\partial R(q)_{32}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix}^\top \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right) \\
&= \mathrm{Tr}\left(\begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{31}} \\ \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{32}} \\ \frac{\partial f}{\partial R(q)_{13}} & \frac{\partial f}{\partial R(q)_{23}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix} \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right) \\
&= \frac{\partial f}{\partial R(q)_{11}} \frac{\partial R(q)_{11}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{21}} \frac{\partial R(q)_{21}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{31}} \frac{\partial R(q)_{31}}{\partial q_c} \\
&\quad + \frac{\partial f}{\partial R(q)_{12}} \frac{\partial R(q)_{12}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{22}} \frac{\partial R(q)_{22}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{32}} \frac{\partial R(q)_{32}}{\partial q_c} \\
&\quad + \frac{\partial f}{\partial R(q)_{13}} \frac{\partial R(q)_{13}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{23}} \frac{\partial R(q)_{23}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{33}} \frac{\partial R(q)_{33}}{\partial q_c} \\
&= \frac{\partial f}{\partial q_c}.
\end{aligned}$$

B. Toy Euclidean Problem

Manifold: 3-dimensional Euclidean space,

$$\mathcal{M}_i = \mathbb{R}^3.$$

Objective function:

$$f(\mathcal{D}) = \sum_{(i,j)\in E} \left\| x_j - x_i - \tilde{d}_{ij} \right\|^2_{R_{ij}^{-1}},$$

where $E \in [N]^2$ is a set of directed measurement edges and $\tilde{d}_{ij}$ is a noisy measurement given by

$$\tilde{d}_{ij} = x_j - x_i + v_{ij},$$

with zero-mean Gaussian measurement noise $v_{ij} \in \mathcal{N}(\mathbf{0}^3, R_{ij})$ and covariance matrix $R_{ij} = R_{ij}^\top \succ 0$.

Gradient:

$$\begin{aligned}
f(\mathcal{D}) &= \sum_{(i,j)\in E} \left\| x_j - x_i - \tilde{d}_{ij} \right\|^2_{R_{ij}^{-1}} \\
&= \sum_{(i,j)\in E} (x_j - x_i - \tilde{d}_{ij})^\top R_{ij}^{-1} (x_j - x_i - \tilde{d}_{ij}).
\end{aligned}$$

The gradient of f with respect to $x_i \in \mathcal{M}_i$ is given by

$$\mathrm{grad}_{x_i} f(x) = -2 \sum_{j:(i,j)\in E} R_{ij}^{-1}(x_j - x_i - \tilde{d}_{ij}) + 2 \sum_{j:(j,i)\in E} R_{ji}^{-1}(x_i - x_j - \tilde{d}_{ji}).$$

C. Rotation Averaging

Manifold: Unit Quaternion,

$$\mathcal{M}_i = S^3.$$

Objective function:

$$f(x) = \sum_{(i,j)\in E} \left\| R(x_i)^\top R(x_j) - \tilde{R}_{ij} \right\|^2_F,$$

where $E \in [N]^2$ is a set of directed measurement edges and $\tilde{R}_{ij}$ is a noisy measurement given by

$$\tilde{R}_{ij} = R(x_i)^\top R(x_j) V_{ij},$$

where V_{ij} is a perturbation rotation matrix corresponding to the rotation about a random axis by a zero-mean Gaussian angle.

Gradient: Let $\bar{f}$ be the smooth extension of f to $\mathbb{R}^4$, given by

$$\bar{f}(x) = \sum_{(i,j)\in E} \left\| R(x_i)^\top R(x_j) - \tilde{R}_{ij} \right\|_F^2.$$

Define $R_i, R_j \in SO(3)$ via

$$R_i = R(x_i)$$
$$R_j = R(x_j).$$

Plug in R_i for all $i \in [N]$ into $\bar{f}$. Note that for all $i \in [N]$, the variable x_i only appears through R_i. As such, we define $\tilde{f}$ in terms of R_i via

$$\tilde{f}(\{R_1, \ldots, R_N\}) = \sum_{(i,j)\in E} \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2.$$

Choose $i \in [N]$ and choose a matrix $V_i \in \mathbb{R}^{3\times 3}$. Define $V = \{\mathbf{0}, \ldots, V_i, \ldots, \mathbf{0}\}$ Then,

$$
\begin{aligned}
D\tilde{f}(\{R_1,\ldots,R_N\})[V] &= \lim_{t\to 0} \frac{\tilde{f}(\{R_1,\ldots,R_i+tV_i,\ldots,R_N\}) - \tilde{f}(\{R_1,\ldots,R_N\})}{t} \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\sum_{j:(i,j)\in E} \left(\left\| (R_i+tV_i)^\top R_j - \tilde{R}_{ij} \right\|_F^2 - \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2 \right) \\
&\quad + \sum_{j:(j,i)\in E} \left(\left\| R_j^\top (R_i+tV_i) - \tilde{R}_{ji} \right\|_F^2 - \left\| R_j^\top R_i - \tilde{R}_{ji} \right\|_F^2 \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\sum_{j:(i,j)\in E} \left(\left\| R_i^\top R_j - \tilde{R}_{ij} + tV_i^\top R_j \right\|_F^2 - \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2 \right) \\
&\quad + \sum_{j:(j,i)\in E} \left(\left\| R_j^\top R_i - \tilde{R}_{ji} + tR_j^\top V_i \right\|_F^2 - \left\| R_j^\top R_i - \tilde{R}_{ji} \right\|_F^2 \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\sum_{j:(i,j)\in E} \mathrm{Tr}\left(2\left(tV_i^\top R_j\right)^\top \left(R_i^\top R_j - \tilde{R}_{ij} \right) + \left(tV_i^\top R_j\right)^\top \left(tV_i^\top R_j\right) \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left(2\left(tR_j^\top V_i\right)^\top \left(R_j^\top R_i - \tilde{R}_{ji} \right) + \left(tR_j^\top V_i\right)^\top \left(tR_j^\top V_i\right) \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\sum_{j:(i,j)\in E} \mathrm{Tr}\left(2tR_j^\top V_i R_i^\top R_j - 2tR_j^\top V_i \tilde{R}_{ij} + t^2 R_j^\top V_i V_i^\top R_j \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left(2tV_i^\top R_j R_j^\top R_i - 2tV_i^\top R_j \tilde{R}_{ji} + t^2 V_i^\top R_j R_j^\top V_i \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\sum_{j:(i,j)\in E} \mathrm{Tr}\left(2tV_i R_i^\top - 2tR_j^\top V_i \tilde{R}_{ij} + t^2 V_i V_i^\top \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left(2tV_i^\top R_i - 2tV_i^\top R_j \tilde{R}_{ji} + t^2 V_i^\top V_i \right) \Bigg) \\
&= \lim_{t\to 0} \Bigg(\sum_{j:(i,j)\in E} \mathrm{Tr}\left(2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} + tV_i V_i^\top \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left(2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} + tV_i^\top V_i \right) \Bigg) \\
&= \sum_{j:(i,j)\in E} \mathrm{Tr}\left(2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} \right) + \sum_{j:(j,i)\in E} \mathrm{Tr}\left(2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \\
&= \mathrm{Tr}\left(\sum_{j:(i,j)\in E} \left(2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} \right) + \sum_{j:(j,i)\in E} \left(2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \right) \\
&= \mathrm{Tr}\left(\sum_{j:(i,j)\in E} \left(2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left(2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \right) \\
&= \mathrm{Tr}\left(V_i^\top \left(\sum_{j:(i,j)\in E} \left(2R_i - 2R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left(2R_i - 2R_j \tilde{R}_{ji} \right) \right) \right) \\
&= \mathrm{Tr}\left(V_i^\top \nabla_{R_i} \tilde{f} \right)
\end{aligned}
$$

Thus, we know

$$
\nabla_{R_i} \tilde{f}(\{R_1,\ldots,R_N\}) = \sum_{j:(i,j)\in E} \left(2R_i - 2R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left(2R_i - 2R_j \tilde{R}_{ji} \right).
$$

Equivalently, we know

$$\nabla_{R(x_i)}\bar{f}(x) = \sum_{j:(i,j)\in E}\left(2R(x_i) - 2R(x_j)\tilde{R}_{ij}^\top\right) + \sum_{j:(j,i)\in E}\left(2R(x_i) - 2R(x_j)\tilde{R}_{ji}\right).$$

Finally, we again use chain rule, according to

$$\nabla_{x_i}\bar{f}(x) = \begin{bmatrix} \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_w}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_x}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_y}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_z}\rangle_F \end{bmatrix}$$

with

$$\text{grad}_{x_i} f = P_{x_i}(\nabla_{x_i}\bar{f}).$$

D. Rotation-Invariant Formation Control

Agent i's manifold is given by

$$\mathcal{M}_i = \mathbb{R}^3 \times S^3.$$

Relevant variables for agent i:

- Decision variable: $x_i = (t_i, q_i) \in (\mathbb{R}^3 \times S^3) \in \mathcal{M}_i$
- Prior position: $\bar{t}_i \in \mathbb{R}^3$
- Prior orientation: $\bar{R}_i \in SO(3)$
- Relative offset to agent j in local reference frame: $\tilde{t}_{ij}$

Objective function:

$$f(x) = \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R(q_i)^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2.$$

Gradient with respect to t_i:

$$\begin{aligned} f(x) &= \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R(q_i)^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2 \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left((t_j - t_i)^\top R(q_i)R(q_i)^\top(t_j - t_i) - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left((t_j - t_i)^\top(t_j - t_i) - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left(t_j^\top t_j - 2t_i^\top t_j + t_i^\top t_i - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \end{aligned}$$

$$\text{grad}_{t_i} f(x) = 2\alpha(t_i - \bar{t}_i) + 2(1-\alpha)\sum_{j:(i,j)\in E}(t_i - t_j + R(q_i)\tilde{t}_{ij}) + 2(1-\alpha)\sum_{j:(j,i)\in E}(t_i - t_j - R(q_j)\tilde{t}_{ji})$$

Gradient with respect to q_i:

First, define $\bar{f}$ as smooth extension of f, where $\mathcal{M}_i$ is extended to $(\mathbb{R}^3 \times \mathbb{R}^4)$. Additionally, define $R_i = R(q_i)$ and define $\tilde{f}$ as $\bar{f}$ with all instances of $R(q_i)$ replaced with R_i, given by

$$\tilde{f}(\{t_i, R_i\}_{i\in[N]}) = \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R_i - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R_i^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2.$$

Choose a matrix $V_i \in \mathbb{R}^{3\times 3}$, and define $V = \{(\mathbf{0}^3, \mathbf{0}^{3\times 3})^{i-1}, (\mathbf{0}, V_i), (\mathbf{0}^3, \mathbf{0}^{3\times 3})^{N-i}\}$. Then,

$$
\begin{aligned}
D\tilde{f}(\{t_i, R_i\}_{i\in[N]})[V] &= \lim_{t\to 0} \frac{\tilde{f}((t_1, R_1), \ldots, (t_i, R_i + tV_i), \ldots, (t_N, R_N)) - \tilde{f}(\{t_i, R_i\}_{i\in[N]})}{t} \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \left\|R_i + tV_i - \bar{R}_i\right\|_F^2 - \alpha \left\|R_i - \bar{R}_i\right\|_F^2 \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(\left\|(R_i + tV_i)^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2 - \left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2\right)\Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \left\|R_i - \bar{R}_i + tV_i\right\|_F^2 - \alpha \left\|R_i - \bar{R}_i\right\|_F^2 \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(\left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij} + tV_i^\top (t_j - t_i)\right\|_2^2 - \left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2\right)\Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \mathrm{Tr}\left(2tV_i^\top (R_i - \bar{R}_i) + t^2 V_i^\top V_i\right) \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(2t(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) + t^2 (t_j - t_i)^\top V_i V_i^\top (t_j - t_i)\right)\Bigg) \\
&= \lim_{t\to 0} \Bigg(\alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i) + tV_i^\top V_i\right) \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) + t(t_j - t_i)^\top V_i V_i^\top (t_j - t_i)\right)\Bigg) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \sum_{j:(i,j)\in E} 2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \mathrm{Tr}\left(\sum_{j:(i,j)\in E} 2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i)\right) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \mathrm{Tr}\left(\sum_{j:(i,j)\in E} 2V_i^\top (t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right) \\
&= \mathrm{Tr}\left(2\alpha V_i^\top (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2V_i^\top (t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right) \\
&= \mathrm{Tr}\left(V_i^\top \left(2\alpha (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right)\right) \\
&= \mathrm{Tr}\left(V_i^\top \nabla_{R_i} \tilde{f}\right)
\end{aligned}
$$

Thus, we know

$$\nabla_{R_i} \tilde{f}(\{t_i, R_i\}_{i\in[N]}) = 2\alpha (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top.$$

Equivalently, we know

$$\nabla_{R(q_i)} \bar{f}(x) = 2\alpha (R(q_i) - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R(q_i)^\top (t_j - t_i) - \tilde{t}_{ij})^\top.$$

Finally, we again use chain rule, according to

$$\nabla_{q_i} \bar{f}(x) = \begin{bmatrix} \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iw}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{ix}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iy}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iz}} \rangle_F \end{bmatrix}$$

with

$$\mathrm{grad}_{q_i} f = P_{q_i}(\nabla_{q_i} \bar{f}).$$

E. Toy Problem

Goal: Find a problem with manifolds $\mathcal{M}_i$ endowed with Euclidean metric $\langle \cdot, \cdot \rangle$ and a function $f(x)$ such that grad$f(x)$ is L-Lipschitz.

Manifolds:

- $\mathcal{M}_1 = \mathbb{R}^{n_1}$, n_1-dimensional Euclidean space
- $\mathcal{M}_2 = S^{n_2-1} \subset \mathbb{R}^{n_2}$, sphere in n_2-dimensional space
- $\mathcal{M}_3$

F. PGO

Let each manifold $\mathcal{M}_i$ be a product manifold given by

$$\mathcal{M}_i = \mathbb{R}^3 \times S^3.$$

A pose on $\mathcal{M}_i$ is denoted $X_i = (P_i, q_i)$.

Consider a measurement graph $G = (V, E)$ with vertices $V = [N]$ and edges E. If $(i, j) \in E$, then agent i takes a measurement